Nagar Yuwak Shikshan Sanstha's
Yeshwantrao Chavan College of Engineering
(An Autonomous Institution affiliated to Rashtrasant Tukadoji Maharaj Nagpur University) Hingna Road, Wanadongri, Nagpur - 441 110
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~~Department of Computer Technology~~

~~**Vision of the Department**~~

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society. **Mission of the Department**

To nurture computer engineering professionals by offering: Dynamic learning environment that encourages problem-solving using emerging technology, · National Education Policy aligned evidence-based technical education, fostering innovation, research, and industry–institute collaboration, Holistic, ethical and sustainable development for lifelong success.

Session 2026-2027(ODD)

Vision: Dream of where you want.	**Mission:** Means to achieve Vision

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	**Preparation**	**P: Preparation**	**Pep-CL abbreviation pronounce as Pep-si-lL easy to recall**
PEO2	**Core Competence**	**E: Environment (Learning Environment)**	
PEO3	**Breadth**	**P: Professionalism**	
PEO4	**Professionalism**	**C: Core Competence**	
PEO5	**Learning Environment**	**L: Breadth (Learning in diverse areas)**	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:
Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety. 

Srushti Chaudhari – 30-9-26
Name and Signature of Student and Date

(Signature and Date in Handwritten)

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Session	2026-27 (ODD)	Course Name	LAB: Software Engineering
Semester	7th	Course Code	23CT1704
Roll No	24	Name of Student	Srushti Chaudhari

Practical Number	8
Course Outcome	Upon successful completion of the course the students will be able to; 1. Choose appropriate software engineering process model, requirement engineering principles and software designing fundamentals for a given project. 2. Select appropriate testing strategy and apply testing principles for testing a given application. 3. Apply basics of software configuration management, version control and change control in software development. 4. Evaluate cost estimation, effort and severity of software risk for given application. 5. Perform basic operations on Sub-version for software version control.
Aim	To write a program to find out the estimation of cost and effort using the COCOMO (Constructive Cost Model).
Problem Definition	To develop a C program that estimates the effort, development time, average staff size, productivity, and total development cost of a software project using the Basic COCOMO model. The estimation is based on the project size in KLOC (Kilo Lines of Code) and the selected project type: Organic, Semi-Detached, or Embedded.

Theory (100 words)	COCOMO stands for Constructive Cost Model. It was developed by Barry W. Boehm in 1981. It is used to estimate the effort, development time, staff requirement, and cost of a software project. The main input to Basic COCOMO is the project size measured in KLOC (Kilo Lines of Code). COCOMO classifies software projects into three modes: **Mode Description** **Organic** Small project with experienced team and well-understood requireme **Semi-Detached** Medium-size project with mixed team experience and complexity **Embedded** Complex project with strict hardware/software constraints **COCOMO Formulae** **1. Effort:** $$E = a \times (KLOC)^{b}$$ where E is effort in Person-Months. **2. Development Time:** $$D = c \times (E)^{d}$$

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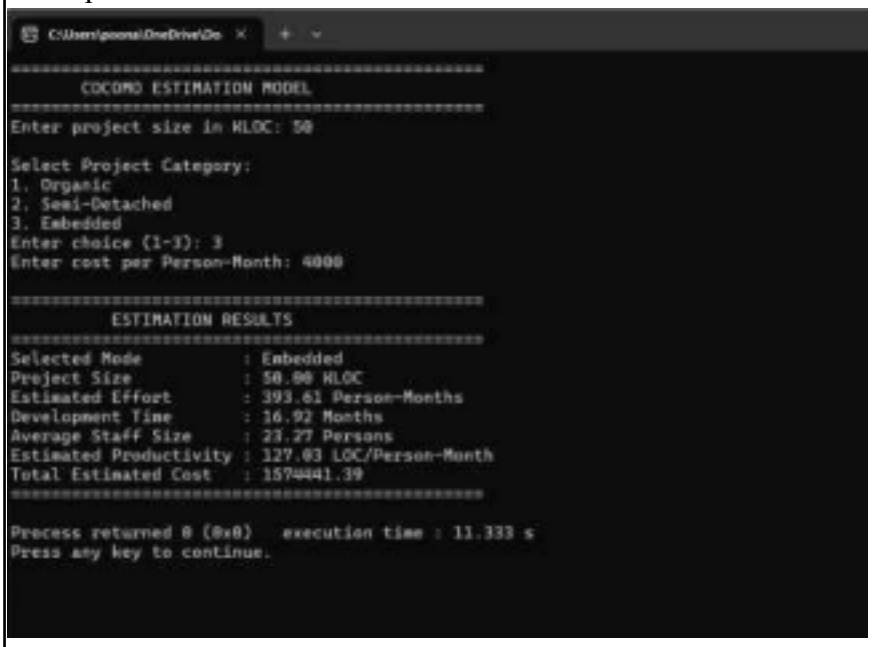
	where D is development time in months. **3. Average Staff Size:** $$Staff = \frac{E}{D}$$ **4. Productivity:** $$Productivity = \frac{KLOC \times 1000}{E}$$ **5. Total Cost:** $$Total\ Cost = E \times Cost\ per\ Person\text{-}Month$$ **Basic COCOMO Constants** **Project Mode a b c d** Organic 2. 1.0 2. 0.3 4 5 5 8 Semi-Detached 3. 1.1 2. 0.3 0 2 5 5 Embedded 3. 1.2 2. 0.3 6 0 5 2
Procedure and Execution (100 Words)	Algorithm(Steps): 1. Start 2. Define the COCOMO constants for all three project modes. 3. Read KLOC. 4. Read the project mode. 5. Read cost per Person-Month. 6. Select the corresponding values of a, b, c, and d. 7. Calculate: 8. Effort = a × KLOC^b 9. Development Time = c × Effort^d 10. Staff Size = Effort / Development Time 11. Productivity = (KLOC × 1000) / Effort 12. Total Cost = Effort × Cost per Person-Month 13. Display the results.

	14. Stop

	Code: #include <stdio.h>
	#include <math.h>
	typedef struct {
	double a;
	double b;
	double c;
	double d;
	} COCOMO_Constants;
	int main()
	{
	double kloc, costPerPM;
	int modeChoice;
	COCOMO_Constants modes[3] = {
	{2.4, 1.05, 2.5, 0.38}, // Organic
	{3.0, 1.12, 2.5, 0.35}, // Semi-Detached
	{3.6, 1.20, 2.5, 0.32} // Embedded
	};
	char *modeNames[3] = {
	"Organic",
	"Semi-Detached",
	"Embedded"
	};

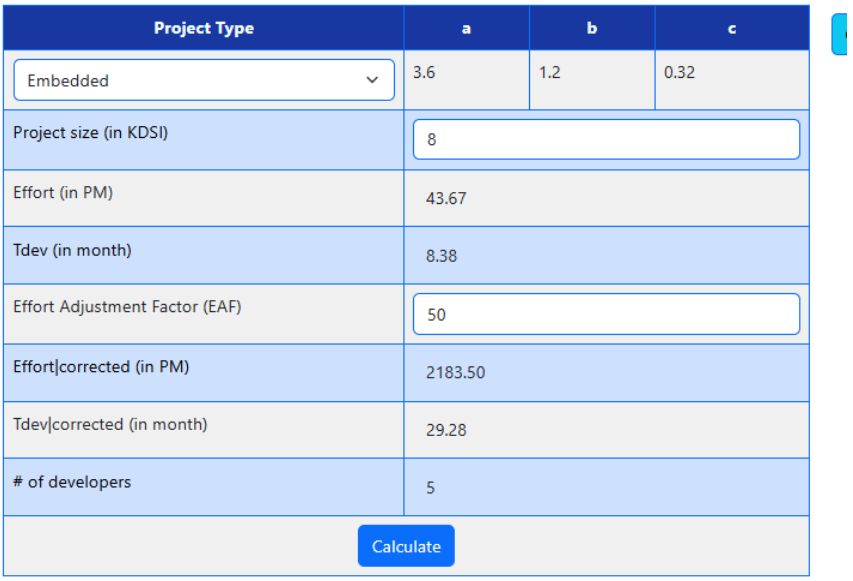
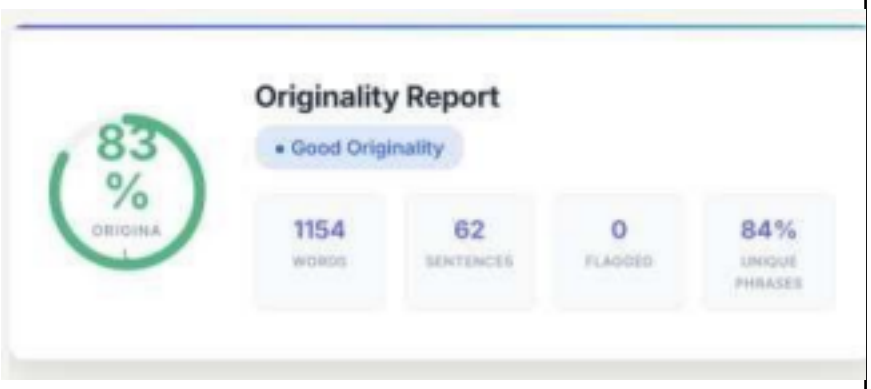
```
printf("=========================================
=== ====\n");
   printf(" COCOMO ESTIMATION MODEL\n");
printf("=========================================
==== ====\n");
    // Input projectsize
    printf("Enter project size in KLOC: ");
    scanf("%lf", &kloc);
    if (kloc <= 0)
    {
       printf("Invalid KLOC value!\n");
       return 1;
    }
    // Select project mode
    printf("\nSelect Project Category:\n");
    printf("1. Organic\n");
    printf("2. Semi-Detached\n");
    printf("3. Embedded\n");
    printf("Enter choice (1-3): ");
    scanf("%d", &modeChoice);
    if (modeChoice < 1 || modeChoice > 3)
    {
       printf("Invalid category selection!\n");
       return 1;
    }
    // Input cost
    printf("Enter cost per Person-Month: ");
    scanf("%lf", &costPerPM);
    int index = modeChoice - 1;
    COCOMO_Constantsselected = modes[index];
    // COCOMO calculations
    double effort = selected.a * pow(kloc, selected.b);
    double devTime = selected.c * pow(effort,
    selected.d);  double staffSize = effort / devTime;
    double productivity = (kloc * 1000.0) / effort;
    double totalCost = effort * costPerPM;
   // Display results
printf("\n=========================================
==== =====\n");
   printf(" ESTIMATION RESULTS\n");
printf("=========================================
==== ====\n");
    printf("Selected Mode : %s\n", modeNames[index]);
    printf("Project Size : %.2f KLOC\n", kloc);
    printf("Estimated Effort : %.2f Person-Months\n", effort);
    printf("Development Time : %.2f Months\n", devTime);
    printf("Average Staff Size : %.2f Persons\n", staffSize);
    printf("Estimated Productivity : %.2f
    LOC/Person-Month\n", productivity);
    printf("Total Estimated Cost : %.2f\n", totalCost);
printf("=========================================
==== ===\n");
    return 0; }
```

Output:



Project Type	a	b	c
Organic	2.4	1.05	0.38
Project size (in KDSI)	2		
Effort (in PM)	4.97		
Tdev (in month)	4.60		
Effort Adjustment Factor (EAF)	50		
Effort|corrected (in PM)	248.50		
Tdev|corrected (in month)	20.33		
# of developers	1		
Calculate			

Project Type	a	b	c
Semi-Detached	3	1.12	0.35
Project size (in KDSI)	4		
Effort (in PM)	14.16		
Tdev (in month)	6.32		
Effort Adjustment Factor (EAF)	50		
Effort|corrected (in PM)	708.00		
Tdev|corrected (in month)	24.85		
# of developers	2		
Calculate			

	
Output Analysis	The program successfully estimates the required effort, development time, staff size, productivity, and total cost of a software project. The COCOMO model uses the project size in KLOC and the selected development mode to perform the calculations. Organic, Semi-Detached, and Embedded projects use different coefficients, so the estimated effort and time change according to the selected mode.
Link of student Github profile where lab assignment has been uploaded	https://github.com/SRchaudhari15/Software-Engineering-Practicals-24.git
Conclusion	Thus, the C program successfully implements the Basic COCOMO model to estimate the cost and effort required for software development. It also calculates development time, average staff size, and productivity based on the project size and development mode.
Plag Report (Similarity index < 12%)	
Date	30-9-26